Idin Karuei

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PhD in Computer Science | AI Engineer and Data Scientist | 10+ years of industry experience
Supervised/Unsupervised ML, Video Understanding, Computer Vision, Generative AI, Recommender Systems

WORK EXPERIENCE

Head of Data Science ..... Quickplay 2026.01 – Now
Staff Data Scientist ..... RBC Borealis 2025.05 – 2026.01
Applied Scientist II ..... Amazon 2024.01 – 2024.12
Sr. Staff Data Scientist ..... SecurityScorecard 2021.05 – 2023.07
Sr. Data Scientist ..... Orbis Investments 2017.10 – 2021.04
Data Scientist ..... Crelogix 2015.08 – 2017.08

EDUCATION

PhD: Computer Science ..... University of British Columbia 2014
MSc: Electrical Engineering ..... Concordia University 2005
BSc: Electrical Engineering ..... University of Tehran 2003

KEY PROJECTS AND IMPACT

Smart Video Cropping ..... Quickplay

- Created the smart-crop pipeline for flagship product ***AI Studio***
- Converts landscape video to any aspect ratio, e.g. ***9:16***
- Multi-stage ***object detection***, tracking, target and layout selection, fitting
- Evaluation and diagnostics workflows for perceptual quality

LLM-as-a-Judge Evaluation ..... RBC Borealis

- ***LLM-as-a-judge*** recommendation evaluation for wealth management
- Built a validation framework for the LLM-as-a-judge methods
- AI for wealth management: Next Best Conversation, Meeting Preparation

Semantic Substitute Ad Sourcing ..... Amazon

- Implemented ad sourcing models using ***embeddings*** of products and ads
- Developed ***KNN search*** for finding substitute products based on similarity
- Dialed-up a weblab to measure the impact of different variations
- Estimated annual impact: ***$210 million*** sales, ***$108 million*** ad revenue

Ransomware Risk Scores ..... SecurityScorecard

- Developed a model that predicts the likelihood of Ransomware attacks
- As a new product, it generated more than ***$3 million*** in 2022
- The model has ***97%*** precision and ***76%*** recall
- Automatically retrains on new data to stay accurate & adapt to threats
- Reduced computation cost by ***$216 thousand*** per year

HARD SKILLS

AI & ML

- Predictive Modeling
- Unsupervised Learning
- Recommender Systems
- Deep Neural Networks
- Embedding Models
- KNN Search
- Video Processing
- Object Detection
- ML Architecture
- Generative AI
- LLM as a Judge
- NLP, Entity Recognition
- Feature Engineering

Data Science

- Statistical Data Analysis
- A/B Testing
- Hypothesis Testing
- Experiment Design
- Survival Analysis

AI Engineering

- AI Tooling
- Managing Data Scientists

Tech Stack

- Databricks, Snowflake
- Tableau, AWS

Programming

- **PySpark**
- **Python**
- **SQL**
- **R**, C / C++

AI-Assisted Engineering Infrastructure . . . . . Quickplay

- Reusable ***Cursor*** / ***Claude*** workflows: shared rules, skills, quality gates
- Diagnostics and ***regression systems*** that keep fast engineering reliable
- Human-in-the-loop evaluation and AI team operating models

Leading Scoring Updates and Calibration . . . . . SecurityScorecard

- Designed and implemented the new update and calibration process
- Worked with heads of Sales, Customer Support, and Engineering
- Led data scientists and data engineers through impact analysis and release
- Reduced ungating failure rate from ***50%*** to ***0%***

RiskGraph: Supply Chain Risk Estimation . . . . . SecurityScorecard

- Developed a novel probabilistic graph model for 3rd & 4th party risk estimation
- Utilized a ***Bayesian probabilistic model*** combined with ***dynamic query generation*** to infer risk propagation
- Enabled automated, scalable risk assessments across complex supply chain networks

Parked Domain Classification . . . . . SecurityScorecard

- Developed an ensemble model using active learning to detect parked domains from DNS records
- Crucial to scoring, the main product of the company, it increased the detection rate from ***2%*** to ***70%***

Sparta: Python Library for ETL . . . . . SecurityScorecard

- Developed a Python library to standardize the way we access internal data resources
- The caching functionality increases the speed of transforming and loading data
- It saves development and computation time, reduces ***AWS*** and ***Databricks*** usage, and saves money

Improving Ukraine's Cybersecurity Posture Against Russian Hackers . . . . . Volunteer Work

- Collected and curated cybersecurity vulnerabilities of Ukraine's infrastructure and industries
- Sector analysis of Ukrainian and American targets for the threat-intel team and authorities

NLP and Text Classification . . . . . Orbis Investments

- Developed an entity recognition model of corporations, products, & persons with ***98%*** accuracy
- Developed an emotion detection model in *R* to analyze Glassdoor reviews
  Used to track employee sentiment at Wells Fargo following the ***fake accounts scandal***
- Created a CNN model for tweet sentiment analysis, matching top 3rd-party solutions

Portfolio Performance Analysis . . . . . Orbis Investments

- Analyzed investment portfolio and found features with statistically significant effect
- Developed predictive models based on the portfolio and estimated the magnitude of features
- Developed good/bad stock classifiers with F1-score, accuracy, precision, recall above ***80%***

Investment Research . . . . . Orbis Investments

- Developed a model to predict gaming revenue in Macao based on tourism and footfall
  Forecasts outperformed consensus and proved more accurate
- Developed investment research models for predicting used car sales in China
  Achieved ***94%*** accuracy, outperforming predecessors during the pandemic
- Built a predictive model for retail growth of Dollar General in the US, leading to a sizeable investment

Credit Risk Forecasting . . . . . Crelogix Acceptance Corporation

- Built models to predict loan defaults and early payouts using ***survival analysis*** and compound interest
- Enabled accurate revenue forecasting, credit reserves, and loss mitigation
- Models were audited and validated by PwC